

SOFTWARE ENGINEERING - LAB 3

Architecture Selection:

We chose Layered Architecture for the Smart Lab Equipment C Slot Reservation Portal.

Architectural Choice and Reasons

1. Clear Separation of Responsibilities:

Layered architecture separates the system into different components for user interaction, reservation management, equipment and calibration management, disciplinary management, and database operations. This makes the system easier to understand and maintain.

2. Easier Reservation Management:

The architecture allows reservation-related rules to be handled separately, including checking real-time equipment availability, verifying calibration status, preventing double-booking, and enforcing the maximum 2-hour reservation limit.

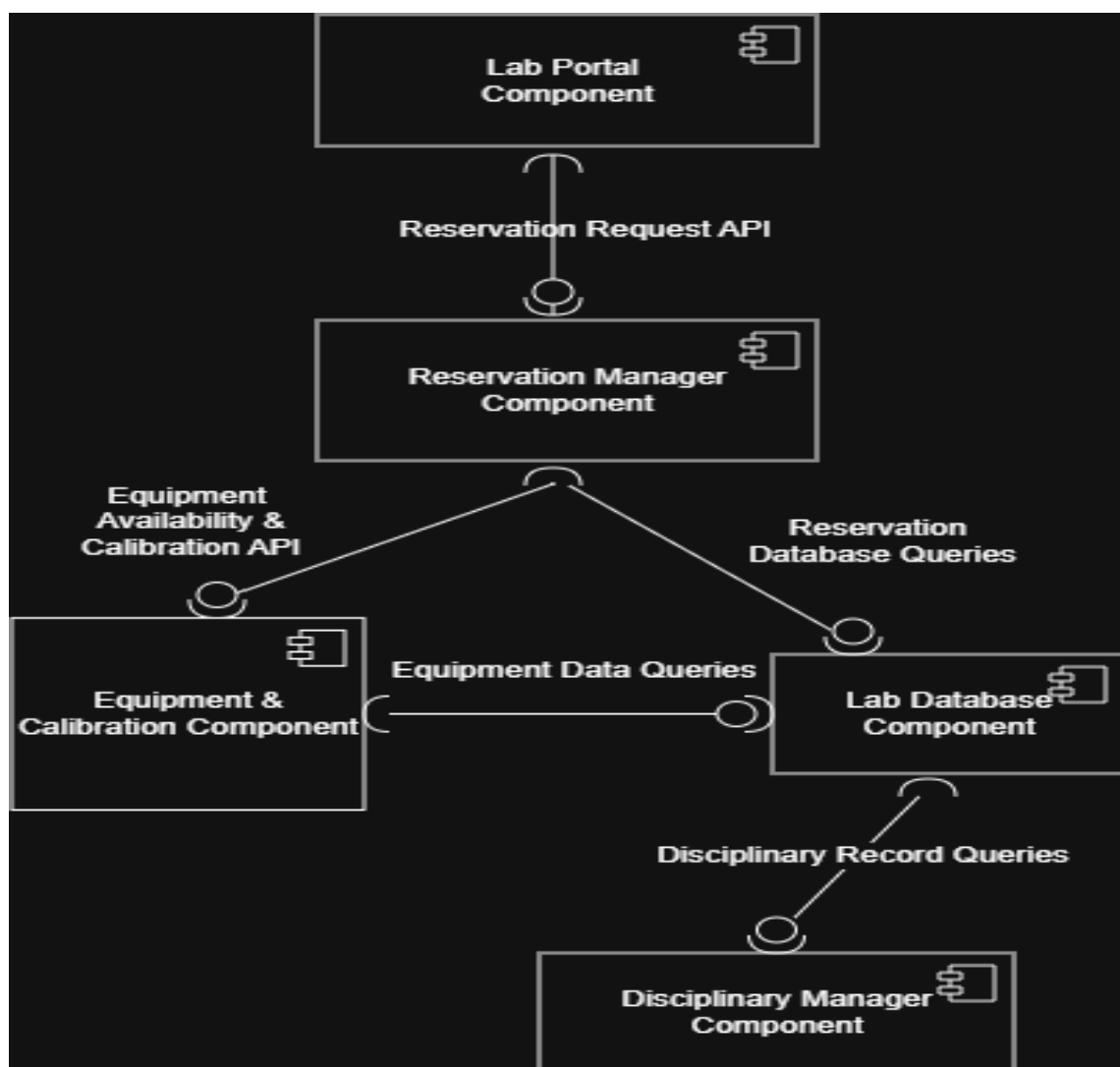
Security Advantage

Layered architecture allows access to sensitive student, reservation, and equipment information to be controlled through

the appropriate system layer. Database access can be restricted to authorized components, improving security.

Performance Benefit

The reservation management component can use database-level locking when processing concurrent reservation requests. This helps prevent race conditions and double-booking while supporting the required **200 ms processing time**.



Component Diagram Description:

The component diagram represents the architecture of the **Smart Lab Equipment & Slot Reservation Portal** using five main components: **Lab Portal**, **Reservation Manager**, **Equipment & Calibration**, **Lab Database**, and **Disciplinary Manager**.

- **Lab Portal Component:** Provides the interface through which students and lab technicians interact with the system and submit reservation requests.
- **Reservation Manager Component:** Handles equipment reservations, checks reservation requests, and manages reservation-related operations.
- **Equipment & Calibration Component:** Provides equipment availability and calibration information required before making a reservation.
- **Lab Database Component:** Stores and retrieves reservation, equipment, calibration, and disciplinary information.
- **Disciplinary Manager Component:** Handles disciplinary records related to late equipment returns.